# FLORENT TARIOLLE

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*Seeking a six-month research internship abroad, starting April 2027.*

## Education

**INSA Rouen Normandy** 2023 – 2027
*Engineering degree, Computer Science and Artificial Intelligence; expected 2027*

Rouen, France. Approx. U.S.-equivalent GPA: 3.7/4.0 overall; 4.0/4.0 latest semester.

**University of Rouen Normandy** 2026 – 2027
*Master's degree, Machine Learning for Artificial Intelligence; expected 2027*

**Lycée Dupuy de Lôme** 2021 – 2023
*CPGE MP — Mathematics and Physics*

Lorient, France. Two-year intensive preparation for competitive entrance examinations to French engineering schools.

## Publications

**DashVMC: Real-Time Discrete World Model Control in Geometry Dash** Sep 2026
*F. Tariolle and F. Yger. PTA workshop, NeurIPS 2026.*

**[arXiv: 2609.40003] [Project page and interactive demo] [Code]**

Developed a world-model agent for live Geometry Dash control from screen pixels. The system combines an $8 \times 8$ FSQ token grid, action-conditioned transformer dynamics, and a lightweight actor-critic initialized by behavioural cloning and refined with PPO entirely in frozen-model rollouts.

Trained offline on approximately two hours of recorded gameplay. Policy refinement improved survival over the cloning baseline on all three evaluated official levels across three controller seeds and on a held-out layout. Decoder-free deployment sustains a 60-FPS capture-to-action loop on an RTX 2060.

**Opportunistic Target Selection: Early Directional Commitment for Query-Efficient Black-Box Adversarial Attacks** May 2026
*F. Tariolle and F. Yger. CAp 2026 (Conférence sur l'Apprentissage automatique).*

**[arXiv: 2605.25663] [Code] [Poster]**

Developed a lightweight wrapper for score-based black-box adversarial attacks. It uses early class-confidence signals to switch from untargeted exploration to a promising target, reducing class drift without gradient access, architectural changes, or prior target-class knowledge.

Evaluated across 4,500 ImageNet attack runs on five classifiers. On ResNet-50, improved attack success by up to 27 percentage points and reduced censored-mean attack iterations by 43%. Presented a poster at CAp 2026.

## Research Projects and Awards

**Finalist, Hack the World(s) Hackathon** Jun 2026
*Geometry-aware joint-embedding EEG · 24-hour JEPA and world-model challenge*

Selected as one of 100 participants from 650+ applicants for a hackathon sponsored by Yann LeCun. Our team finished among the top five of 25 teams.

Built a model that learns from unlabeled EEG recordings. Explored whether training that respects the curved geometry of covariance matrices leads to more structured representations.

**[Code] [Presentation]**

## Experience

### InterDigital

**Generative AI Consultant** Oct 2026 – Present

*Freelance · France, remote*

Advising InterDigital's R&D teams on generative AI workflows, plugins, and skills, building on tools I initiated during my research internship. Helping teams develop and maintain these tools independently through practical guidance on AI-assisted development, reliability, and security.

**Deep Learning Research Intern** May – Sep 2026

*Rennes, France*

Researched learned decoder-side filter selection as an alternative to neural in-loop filtering for VVC (H.266) video decoding, with the aim of reducing compression and quantization artifacts. Authored and filed more than 10 patent applications arising from this research.

### Enedis

**Machine Learning Intern** Jan 2026 – Present

*Part-time capstone project, 25 hours/week · Rouen, France*

Building a digital twin of Enedis's distribution network by mapping synthetic customer load profiles to distribution substations and linking them to primary substations. The project combines customer curves generated by DeepCourboGen with measured substation load data, and includes ongoing work on spatial and temporal adjustments to make the synthetic profiles more realistic for their assigned locations.

*Paused May–September 2026 during the research internship at InterDigital.*

## Open-Source Projects

**VisualTorch** Jul 2026 – Present

*Maintainer and MCP Integration Lead · Official PyTorch Ecosystem project*

Maintaining a library for visualizing PyTorch neural-network architectures. Designed and implemented its MCP integration to generate editable diagrams from model definitions, making architecture inspection available through agent tooling. The project has more than 300 GitHub stars.

**Rose — Real-time customization for League of Legends** Sep – Dec 2025

*Co-founder and Lead Developer · Open source*

Co-founded and co-developed an open-source customization tool; remain the project's top contributor. Built core application components spanning the Python backend, in-client JavaScript plugins, and their real-time WebSocket bridge.

The project has since reached 30,000 daily active users and 90,000 monthly active users, with 700+ GitHub stars and 20+ contributors.

## Technical Skills

**Programming:** Python, C++
**Frameworks:** PyTorch
**Research:** Self-supervised learning, world models, model-based reinforcement learning and planning
**Languages:** French (native), English (TOEIC 970/990), Spanish (intermediate)

## Interests

League of Legends (top 0.1% of ranked players; two LAN wins), bouldering, and strength training.

## References

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